

Dental Significance of Neurologic Disorders

OMFS: 512 Biomedical and Integrated Clinical Sciences

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

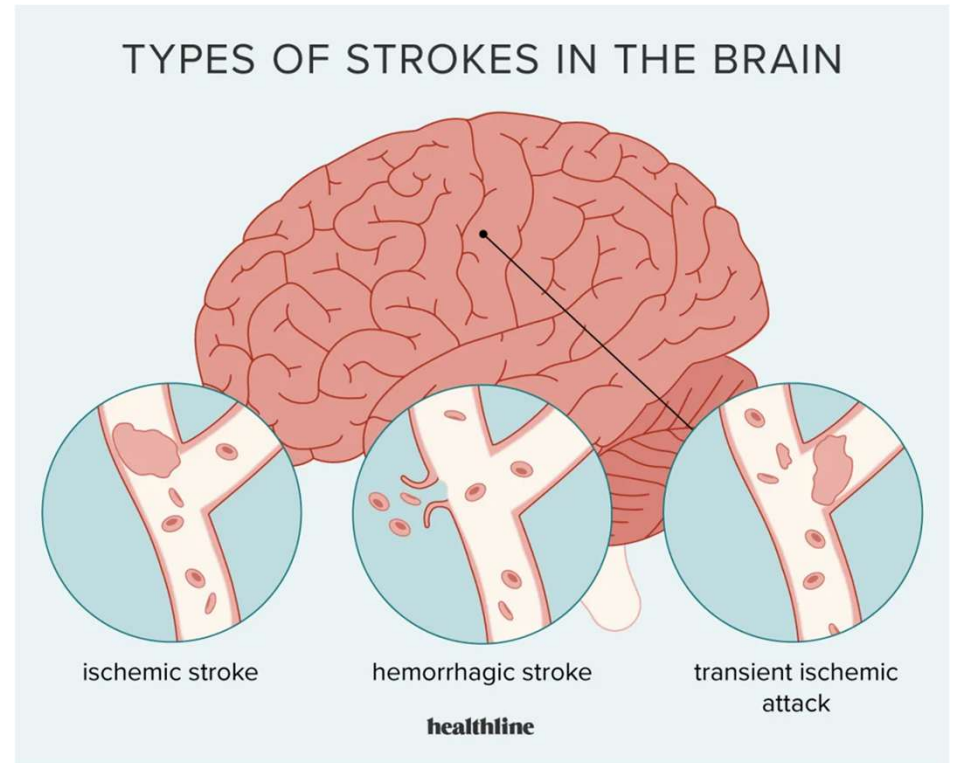
1. Review the causes and clinical manifestations of various neurologic disorders.
2. Learn how each of the neurologic disorders can impact dental treatment.
3. Discuss the process of obtaining consent for dental treatment in patients experiencing cognitive decline.

Competency statements met

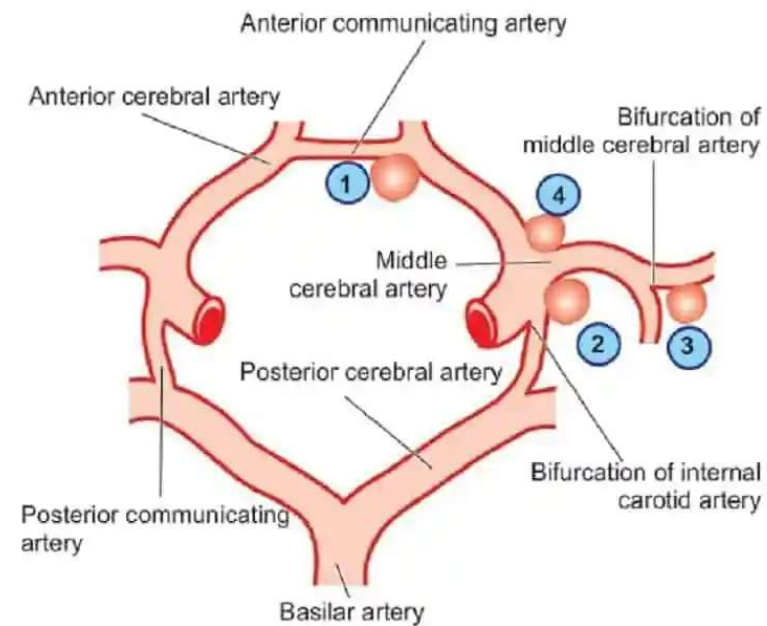
1.3, 1.5, and 2.1

Stroke

- Sudden disruption of oxygen to the brain
- 5th leading cause of death in the United States
- Patient who survive may experience disabilities
 - 10%- no impairment
 - 30 %- mild residual disabilities
 - 15%-30%- disabilities requiring special services
 - 10%-20%- long-term disabilities requiring institutionalization



Subarachnoid Hemorrhage



The circle of Willis showing principal sites of berry (saccular) aneurysms. The serial numbers indicate the frequency of involvement.

<https://medicoapps.org/berry-aneurysm-2/>

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Dental Considerations

- Paralysis
- Medications
 - Anticoagulants
- Dental treatment
 - Treat with compassion
 - Treatment planning
 - Schedule appointment when the patient is alert.
 - Calcified carotid artery atheroma
 - Be careful with local anesthetics containing epinephrine



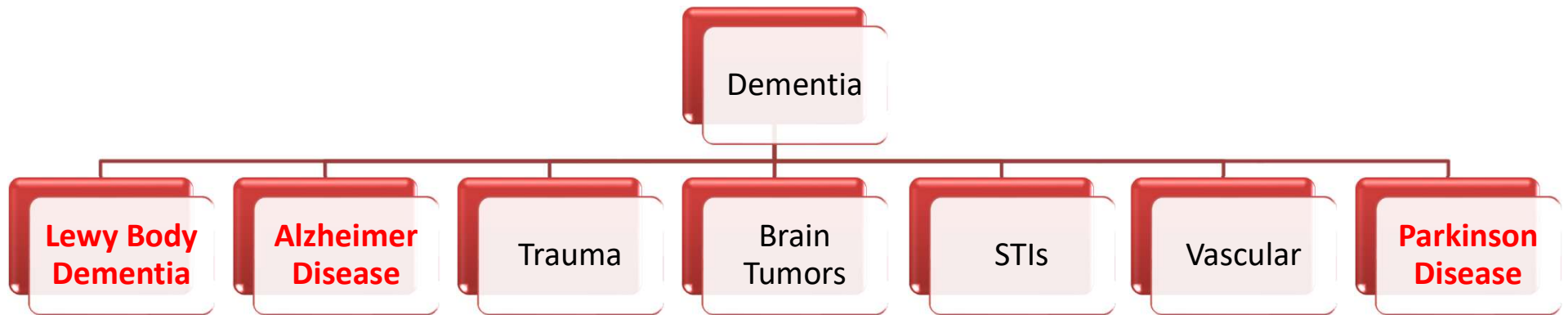
J Am Dent Assoc. 2007 Apr;138(4):483-92.

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TABLE 2. Recognize a Stroke Emergency³

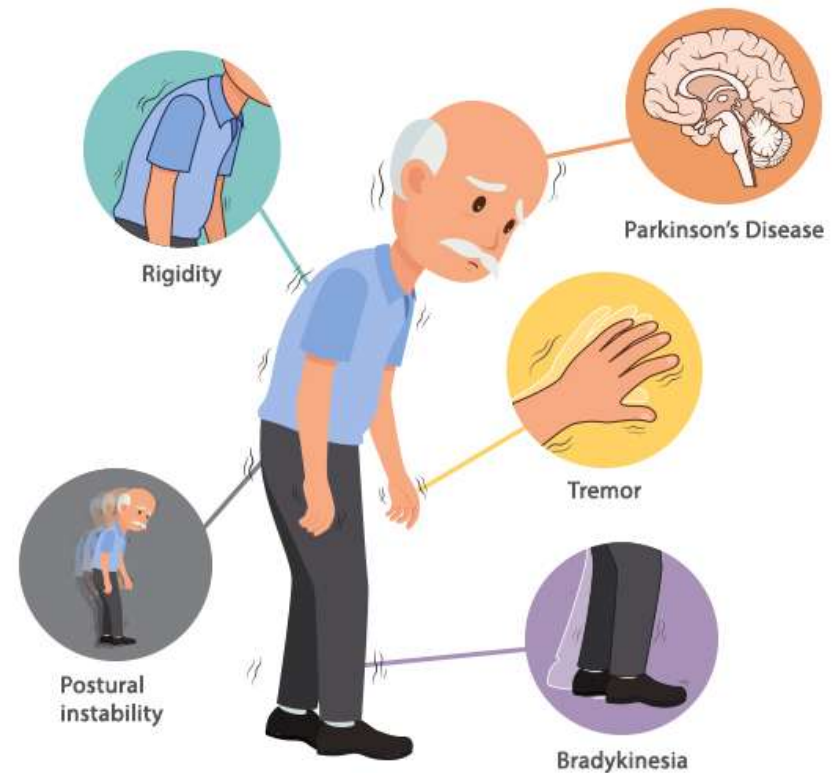
F	FACE	Ask the individual to smile. Drooping on one side of the mouth or face is a sign of stroke.
A	ARMS	Ask the individual to raise both arms. One arm that slowly comes back down or cannot be raised is a sign of a stroke.
S	SPEECH	Ask the individual to repeat a simple sentence. Speech that is slurred or sounds strange is a sign of a stroke.
T	TIME	Call 911 if you see any of these signs. Stroke is an emergency and time is of the essence.

Dementia



Parkinson Disease

- Loss of dopaminergic neurons
- Forms
 - Adult onset
 - Young
 - Juvenile
- Causes
 - Physical damage
 - Chemical exposure
 - Medications



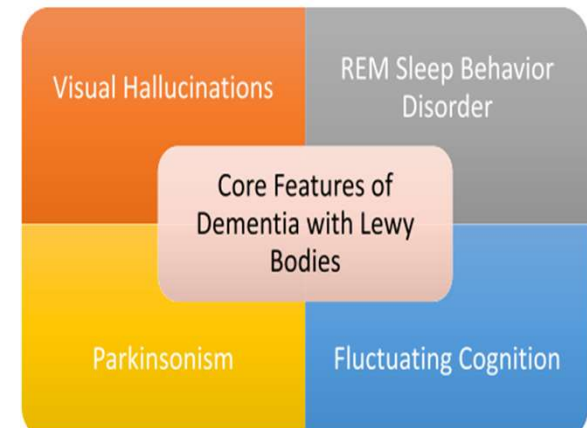
Parkinson Disease

- Frequent dental appointments
- Oral hygiene instructions
- Dental appointments scheduled after medication
 - “Wearing off phenomenon”
- Chair positioning
- Medical drug interactions



Lewy Body Dementia

- Accumulation of misfolded proteins forming “**Lewy bodies**” found in neurons.
- Risk factor
 - Gender
 - Age
 - Genetic disposition
- Sensory, cognitive, and motor dysfunction



Alzheimer Disease

- Deposition of beta-amyloid plaques in the neurons
 - Neurofibrillary tangles

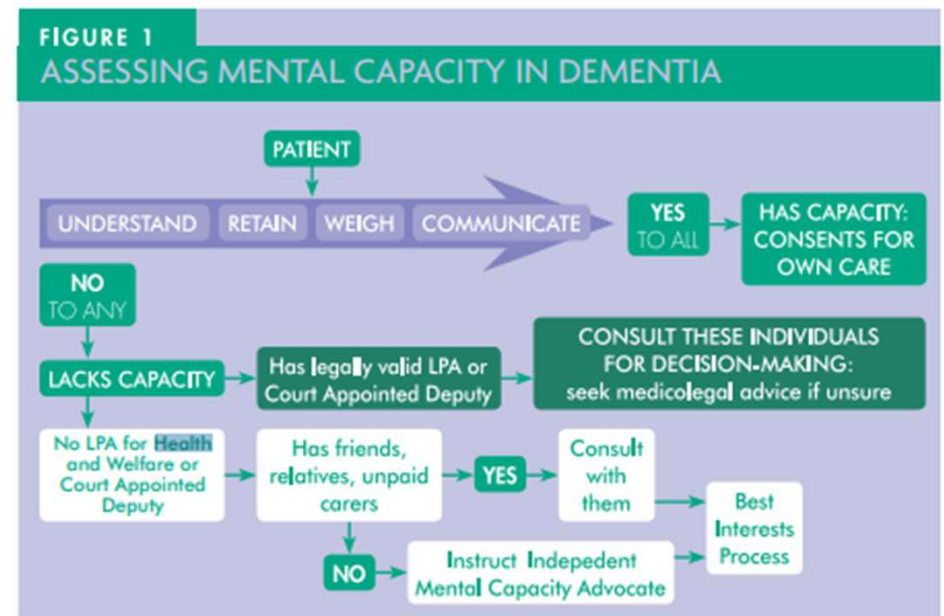
Stages of Alzheimer	
Preclinical	No outward symptoms
Mild/Early	Cognitive impairment; memory loss Mental changes does not interfere with everyday functions
Moderate/Middle	Patients get lost, difficulty with communication, and behavioral changes
Severe/Late	Need assistance with daily activities, difficulty walking and recognizing friends and family Behavioral changes; aggressive behavior

Dementia

- Dental Considerations
 - Xerostomia
 - Positive and supportive communication
 - 3-month recalls
 - Treatment during late stages of disease
 - Avoid complex dental treatment
 - Removal prosthesis may need to be taken away to prevent injury
 - Referral to a dental specialist

Patient Consent

- Determine if the patient is qualified to make rational decisions.
- Permission from the patient's family/guardian
- Discuss the patient's desires for their oral health prior to cognitive decline.
- Power of Attorney



Prim Dent J. 2020;9(2):31-37

Epilepsy

- Group of disorders characterized by changes in neurologic function, altered consciousness, or involuntary movements.
- Triggers:
 - Anxiety
 - Fear
 - Anger
 - Bright lights
 - Alcohol and drug abuse
 - Dehydration and starvation
 - Changes in the type or dosage of antiseizure medications



Epilepsy

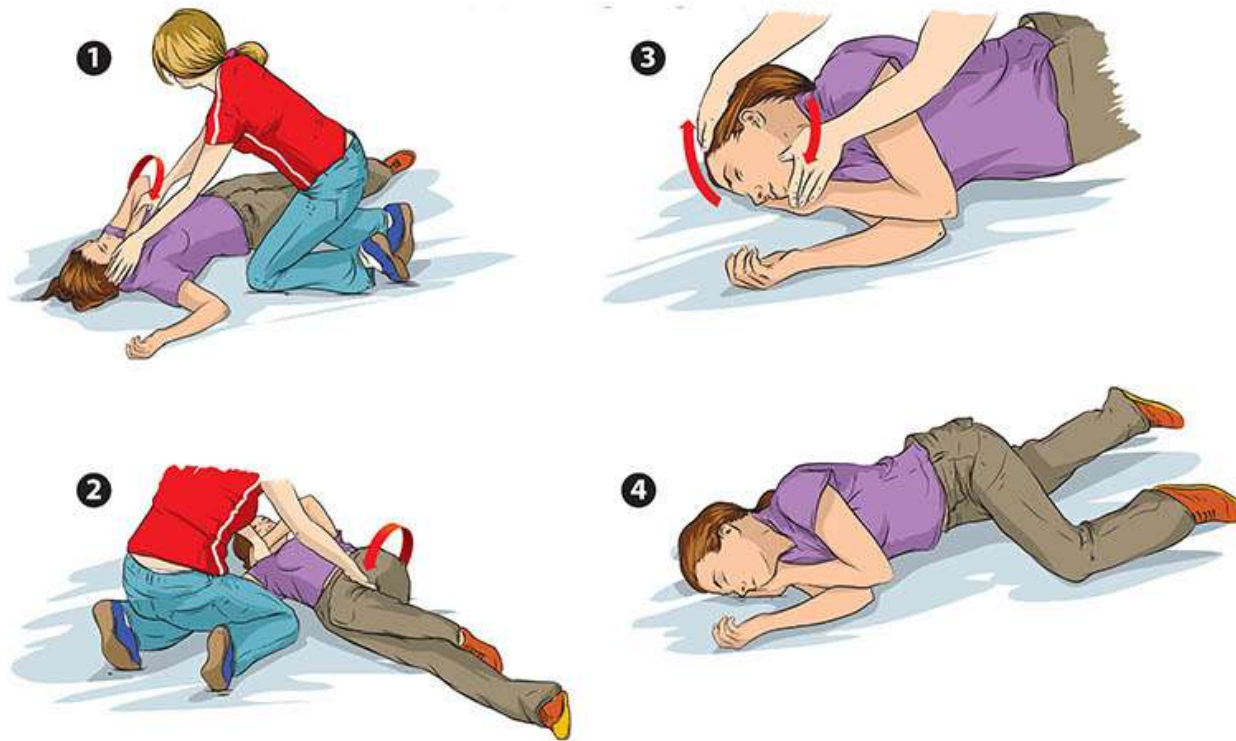
- Oral findings
 - Seizure-induced injury to the hard and soft tissues
 - TMJ dislocation, tooth displacement, soft tissue ulceration, fracturing of teeth and/or dental restorations
 - Medication-related changes to the soft tissues
 - Dilantin (Phenytoin)
 - Valproic acid

Epilepsy



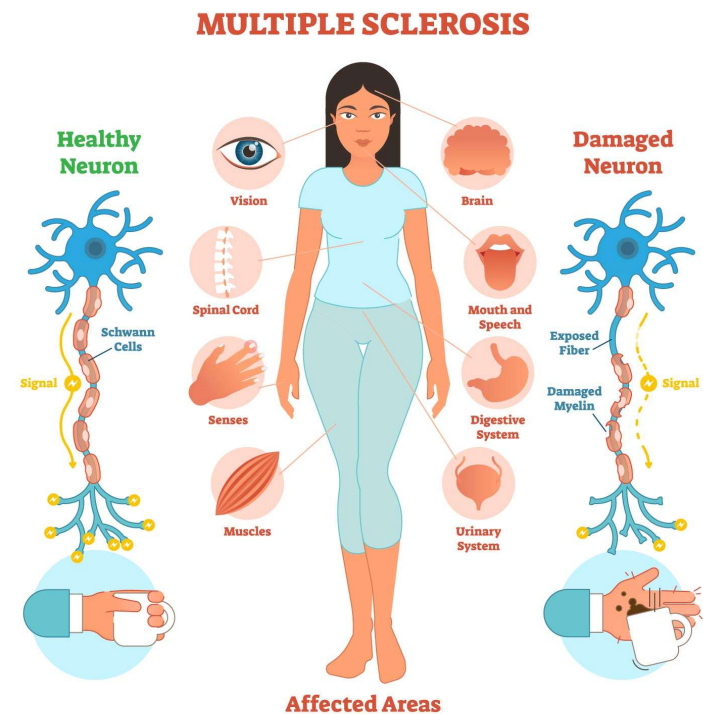
Phenytoin-induced gingival enlargement: a dental awakening for patients with epilepsy Case Reports 2013;2013.

Epilepsy



Multiple Sclerosis (MS)

- Most common autoimmune disease involving the nervous system.
 - Demyelination of the corticospinal tract neurons
 - Muscle atrophy
- Occurs in young adults



shelteringarms.com/rehablog/symptoms-of-multiple-sclerosis

Multiple Sclerosis (MS)

- Dental Considerations
 - Abnormal facial pain
 - Ability to maintain oral health
 - Dental treatment during phases of remission
 - Short morning appointments
 - Wheelchair accommodations
 - Adverse reactions associated with drug therapy
 - Dry/burning mouth
 - Susceptibility to infection

Multiple Sclerosis (MS)

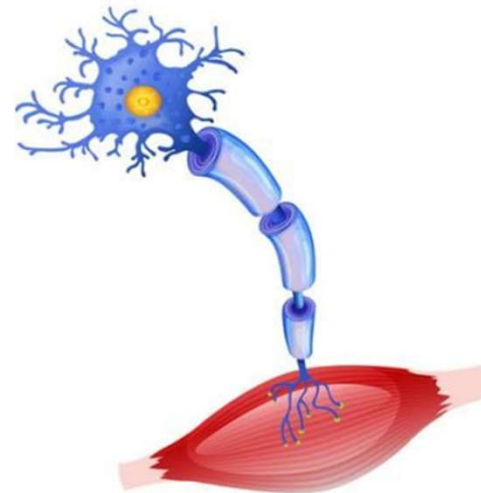


Amyotrophic Lateral Sclerosis (ALS)

- Lou-Gehrig's disease
- Loss of motor neurons in the cerebral cortex, brain stem, and spinal cord.
- Disease occurs between 50-70 years of age
- Life span: 2.5-3 years after the diagnosis

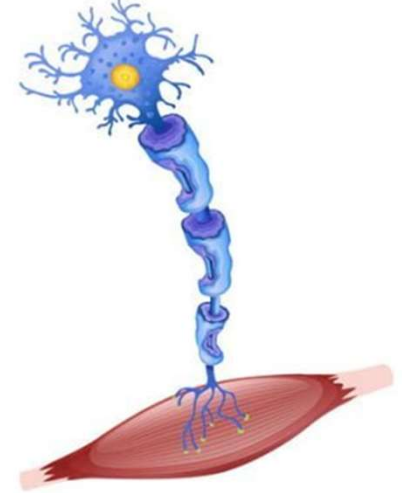
Amyotrophic Lateral Sclerosis (ALS)

normal nerve cell



muscle contracts

nerve with sclerosis



muscle unable to contract

Amyotrophic Lateral Sclerosis (ALS)

- Symptoms
 - Hyperactive reflexes
 - Loss of complex reflexes and dexterity
 - Chronic drooling
 - Issues with speech and swallowing
 - Decrease in tongue size

Amyotrophic Lateral Sclerosis

Table 2. Dental management of ALS.

1. Use of sedative drugs is contraindicated.
 2. Treatment should be administered with the patient in the upright position.
 3. NPO for 12 hours before treatment.
 4. Avoid contact with soft-tissue areas, since this induces gagging.
 5. Frequent breaks in treatment, allowing patient rest.
 6. Involve primary care-giver in oral hygiene procedures.
 7. Use of topical anesthetics is contraindicated.
 8. Use of topical fluoride may induce nausea.
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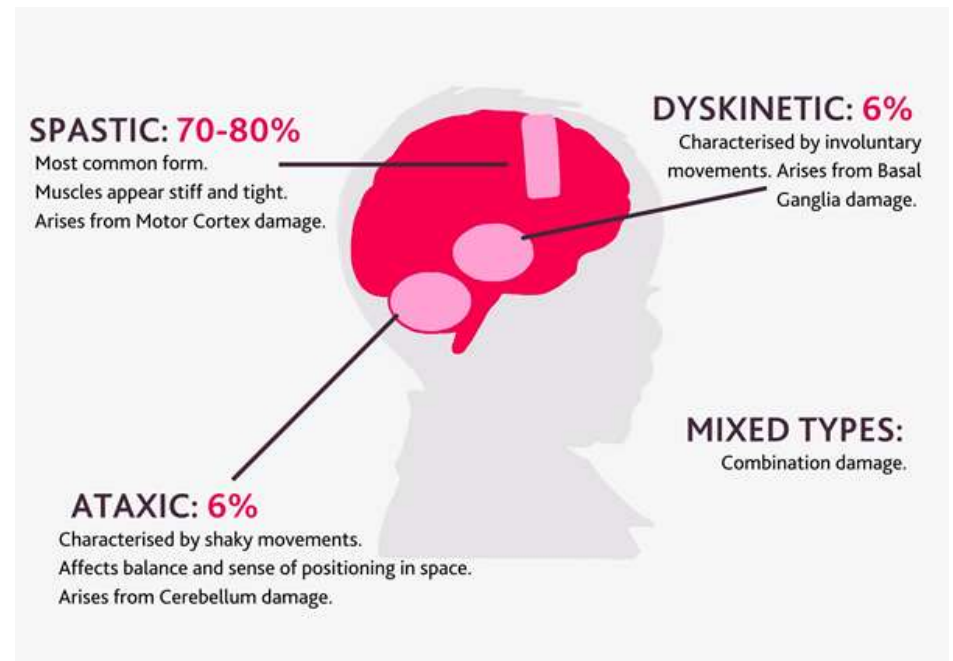
Spec Care Dentist. Nov-Dec 1993;13(6):241-4

ALS VERSUS MS

MS (Multiple Sclerosis) is an autoimmune neurodegenerative disorder	ALS (Amyotrophic Lateral Sclerosis) is a progressive and fatal neurodegenerative disorder
Affects the brain and spinal cord by disrupting the myelin sheath	Affects the neurons in the brain and spinal cord
Can cause sensory disturbances	Only affects motor functions, not sensory functions
Symptoms include easy fatigability, difficulty in limb movements, disturbances in vision, bladder incontinence, numbness, tingling sensation, muscle stiffness, and spasms	Symptoms include progressive muscle weakness ultimately ending up with paralysis, muscle cramping, twitching and slurred speech Visit www.pediaa.com

Cerebral Palsy (CP)

- Common congenital disability associated with damage to the brain.
- Clinical Manifestations
 - Poor motor control
 - Abnormal gait
 - Loss of ability to walk
 - Difficulty with speech, swallowing, and excessive drooling.



Cerebral Palsy Dental Considerations

- Dental attrition and erosion
- Increased salivation
- Increased risk of dental caries
- Issues with manual dexterity
- Gingival hyperplasia
- Wheelchair accommodation
- Patient communication
- Issues with muscle control

Resources

1. Geddis-Regan A, Kerr K, Curl C. The Impact of Dementia On Oral Health And Dental Care, Part 2: Approaching And Planning Treatment. *Prim Dent J*. 2020;9(2): 31-37.
2. Asher RS, Alfred T. Dental management of long-term amyotrophic lateral sclerosis: Case report. *Spec Care Dentist*. Nov-Dec 1993;13(6):241-4.
3. Little JW, Miller CS, Rhodus NL. Neurologic Disorders. Little and Falace's Dental Management of the Medically Compromised Patient. Elsevier. 2018. Chapter 27:516-543.
4. Mupparapu M, Kim IH. *J Am Dent Assoc*. 2007 Apr;138(4):483-92.
5. Scully C. Neurology and Impairment and Disability. Medical Problems in Dentistry 6th Ed. Elsevier. 2010. Chapters 13:305-351 and 28:613-624.
6. Morimoto Y, Hayashi M, Tanaka Y, Mikuzuki L, Nishizaki H. Management for the patients with severe Parkinson's disease during dental treatments and tooth extractions: A retrospective observational study. *J Dent Sci*. 2024;19(1):261-267